RESEARCH TITLE

The aim of this document is to propose future research to understand the possibility to implement a Machine Learning-Based Customer Profiling in the B2B Printing Industry: a comparative evaluation against traditional segmentation methods.

RESEARCH PROBLEM

In today's customer-centric business landscape, effective communication between marketing functions and senior leadership is essential to organisational success. The ability to develop precise and insightful customer profiles constitutes a fundamental pillar of strategic decision-making for organisations seeking sustainable growth and enhanced customer satisfaction (Kasem et al., 2024).

Customer profiling and segmentation is widely used in almost every type of business and sector. It is a paramount instrument to focus efforts on the right customers.

Customer-segmentation profiling equips an organisation with the analytical insight needed to identify which specific audience groups demonstrate the strongest interest or likelihood of engaging with a particular product. (Mohamed et al., 2022).

Artificial Intelligence (abbreviated as AI in this report) has recently supported the customer profiling: from data collection, screening and cleaning to segmentation and clustering, from predictive behavioural analysis to personalization of communication and interaction.

Many sectors have benefited from the usage of AI in customer profiling replacing the traditional profiling techniques, the Fintech sector, for example has widely adopted new AI techniques to improve analysis of risks and returns.

AI-driven optimisation delivers stronger risk-adjusted performance than traditional methods, achieving Sharpe ratios well above industry norms (Sappa, 2024: 182).

The printing industry is historically a very traditional sector where most of the companies are small and medium enterprises, mainly family owned. This background is one of the reasons of the hesitancy in implementing AI in their standard operation and, for example, in the customer profiling.

In a report of the PRINTING United Alliance and NAPCO research (2024: 6) this delay in using AI is reported and one of the highlights is that AI's transformative potential in the printing sector remains largely under-recognised, as many industry professionals are still unfamiliar with its full range of capabilities and practical uses. This gap in understanding creates both an obstacle to adoption and a valuable opening for organisations to invest in education, build internal expertise, and drive innovation across the industry.

This delay can be, anyway, also considered as an important opportunity to modernise the printing business. The AI implementation in the next years in the customer analytics is forecasted to increase a 14.5% (from the current 12.9% to a 27.4%) and this includes tasks like assessing customer feedback, requests, behaviours, and preferences to uncover opportunities for value creation or problem-solving, and detecting customers who may be at risk of churning (PRINTING United Alliance and NAPCO research, 2024: 10).

This new implementation will imply various ethical and legal considerations such as the possible sharing of patented products or sensitive data regarding products, company initiatives, financial information and any other data that could be provided to competitors. Another sensitive issue would be to reassure the employees that this implementation of AI will not be done to reduce the workforce but rather to improve and streamline the operations of the businesses and increase the accuracy, especially when talking about a critical task such as the customer profiling.

Despite the proven effectiveness of ML-based customer segmentation in sectors such as fintech and e-commerce, there is a notable absence of research exploring its application in the B2B printing industry. This represents a significant gap in both academic literature and industry practice.

A possible implementation could be a customer segmentation using the Machine Learning algorithm K-means on a traditional Recency, Frequency and Monetary Value (RFM) analysis and the results produced by the K-means algorithm then validated to identify the most appropriate number of clusters K (Kasem et al., 2024).

From a computing perspective, this research contributes to the evaluation of K-means clustering against traditional RFM analysis in a domain where computational techniques have not been applied such as the B2B printing industry.

This study fills the existing gap by comparing ML-based customer segmentation with traditional RFM approaches, using the B2B printing sector as real business context. The results are expected to advance academic understanding of ML in underexplored industries and support more data-driven decision-making within the printing sector.

RESEARCH QUESTION

Building on the premises outlined in the previous section, the core research question is the following:

“How effectively can machine-learning techniques classify and segment B2B customers compared with traditional profiling methods in the printing industry?”

This question centres on B2B customer profiling, which, even within large multinational firms, still relies heavily on conventional, spreadsheet-based approaches which are, many times, outdated and manually updated.

It addresses a clear gap in existing research, as ML-driven segmentation has seen limited application or assessment within the printing sector.

From a computing perspective, it also provides an opportunity to evaluate algorithms such as K-means clustering in an industry that has yet to engage with AI-enabled transformation.

RESEARCH AIMS

The purpose of this study is to examine how well machine-learning approaches can classify and segment B2B customers in comparison with established profiling techniques within the printing industry. This sector remains largely underexplored in academic research and continues to depend on traditional, non-AI-driven segmentation practices, making it a valuable context for assessing the added effectiveness of ML-based methods.

RESEARCH OBJECTIVES

- *To review critically Machine Learning vs Traditional Customer profiling methods.* It is important to identify approaches, performance metrics and gaps in current research.
- *To identify and prepare a comprehensive B2B customer dataset* for a proper analysis and segmentation: high attention on data quality and preprocessing techniques.
- *To apply traditional RFM segmentation to the dataset* and collect and divide the data based on the customer segmentation.
- *To apply Machine Learning algorithms (K-means for example) to the same dataset* and collect and divide the data as done for the traditional RFM segmentation.
- *To critically compare the results of both the analysis* using pre-determined performance metrics (accuracy of segmentation, quality of the clusters and relevance in term of business).

Segmentation accuracy will be the primary metric for comparison. Additionally, the time efficiency of ML algorithms compared with traditional segmentation methods will be assessed as a secondary measure of practical value.

EXPECTED OUTCOMES

This research is expected to bring insightful contributions to the printing industry, particularly within the B2B environment, as well as to the academic community.

From the business perspective, the main desired outcome is to demonstrate to SMEs operating in the printing sector how machine learning algorithms, when combined with the traditional RFM approach, can significantly improve and accelerate customer profiling and segmentation. Providing the industry with a reliable, replicable, and data-driven framework that can effectively replace the long-standing reliance on Excel spreadsheets may also serve as a practical way to reduce the hesitancy surrounding the adoption of AI in daily operations. This reluctance remains one of the major obstacles to the broader implementation of advanced analytical tools, and offering a structured, easy-to-apply methodology could help lower this barrier.

From the academic perspective, the study aims to fill an important gap in the literature. While machine learning–based customer segmentation has been widely explored in B2C contexts, especially in e-commerce, online trading, and financial services, there is almost no evidence of its application within the B2B environment of the printing business.

By addressing this gap, the research contributes to the growing body of work on AI adoption in traditional manufacturing industries and highlights how data-driven decision-making can be extended to sectors that have historically been significantly slower in embracing digital transformation.

Lastly, from a computing perspective, the study evaluates the performance of K-means clustering in a domain where computational techniques have not been applied. This contributes to a much deeper understanding of how established machine learning methods can be transferred, adapted, and validated in new industrial contexts, offering insights that may be relevant for future cross-sector applications.

LITERATURE REVIEW

For many companies, customer profiling and segmentation have become fundamental components of an effective business strategy. In B2B contexts especially, where relationships tend to be long-term and committed, effective customer profiling is a key instrument for success. Segmentation enables a company to identify which customer groups to target and how to position its products and services most effectively for each one (Cooil et al., 2006). Despite its recognised importance, the adoption of advanced segmentation techniques varies significantly across industries.

The printing sector worldwide is widely regarded as a conservative industry with limited openness to technological innovation, particularly in customer profiling and segmentation, which in many cases still rely on personal relationships and historical connections. Artificial intelligence has emerged as a key enabler of Industry 4.0. From its origins nearly seven decades ago to recent advances in machine learning, AI now provides essential tools that support organisational management and strengthen decision-making. As a result, AI is expected to reshape industrial environments and significantly boost productivity. This shift also affects the global print manufacturing sector, where firms are increasingly urged to reassess their strategies to stay relevant and competitive (Masod et al., 2021).

This research addresses a notable gap in the literature regarding the use of machine learning algorithms for customer profiling and segmentation in the B2B printing industry, given that such techniques are already widely applied in sectors such as e-commerce, fintech, and retail.

This review examines five key areas: *ML techniques for customer segmentation*, *B2B customer analytics*, *traditional versus ML-based methods*, *AI adoption in the printing industry*, and *the ethical considerations surrounding ML-driven profiling*.

ML techniques for customer segmentation.

Few research have been already done about Machine Learning techniques related to customer segmentation. The findings indicate that machine-learning-based segmentation consistently outperforms traditional statistical approaches across most scenarios, particularly

when evaluating overall profit, accounting outcomes, and opportunity costs (Florez-Lopez and Ramon-Jeronimo, 2009: 114).

Moreover, machine learning offers an extensive range of classification and clustering methods, allowing practitioners to select the most suitable approach based on performance for customer-segmentation tasks. These techniques uncover hidden structures in customer behaviour and related attributes, enabling the formation of coherent and homogeneous groups (Behera et al., 2020).

Machine learning techniques are currently supporting the traditional customer profiling techniques like the RFM technique (Recency-Frequency-Monetary) and improving the outcome of the profiling and segmentation process.

Overall, behaviour-driven customer segmentation that combines RFM analysis with machine-learning clustering provides an effective and actionable approach for organisations seeking deeper customer understanding and more targeted marketing strategies. Future studies could enhance this framework by incorporating additional behavioural indicators, such as engagement levels or product affinities, and by examining more sophisticated clustering methods to extract even richer insights (Juhasz, 2025: 173).

B2B customer analytics.

B2B customer profiling and segmentation is a more elaborate and longer process than customer profiling in a business-to-consumer environment (B2C).

Business-to-business customers tend to be more complex, with a more involved purchasing process and higher transaction values. Business marketers generally choose to work with a smaller number of large buyers rather than the broad base targeted in consumer markets. Because business transactions demand multiple decision makers and a higher level of professional purchasing effort than consumer purchases, maintaining strong and efficient relationships with business clients becomes critically important (Hosseini and Shabani, 2015).

The reason why this research is focused on applications of Machine learning techniques on B2B customer profiling and segmentation is because there is an evident gap on usage of AI

on the Business-to-Business environment. Empirical research on AI-enabled B2B marketing remains strikingly limited, and studies that probe the socio-technical dynamics of its adoption are even more scarce (Keegan et al, 2024: 1025).

Also, is pertinent to mention that AI and Machine Learning techniques have been focused more on business-to-consumer environment (B2C) or e-commerce and this is another solid reason to pursue comprehensive research on AI applications on B2B environment.

Mora Cortez, Højbjerg Clarke and Vagn Freytag (2021), in a systematic review of 88 studies on B2B market segmentation revealed that most segmentation scholarship remains centred on consumer markets (B2C), with only 5–10% of articles in top marketing journals addressing B2B settings. This disproportion underscores a long-standing neglect of industrial segmentation research, a gap made even more pronounced by the added intricacies of B2B buying centres and extended decision-making processes.

On the other side, also businesses, despite acknowledging the benefits of AI, are either reluctant or not technically ready to implement it in their daily operations.

While more and more studies acknowledge the potential value of AI in B2B marketing, few attempts have been made to synthesise the literature. This is problematic as the results generated from different studies tend to be limited, and there is a lack of comprehensive management guidelines to enable companies to integrate AI as an enabler for innovation into their B2B marketing (Han et al., 2021).

traditional versus ML-based methods.

Printing business, especially in B2B environment, has widely used the RFM framework to profile and segment its customers. As the printing business is a conservative and traditional business, the RFM framework is still valid nowadays in the B2B environment and very few companies (mainly MNCs) are exploring different ways to improve and optimize the customer profiling and segmentation with the goal of target the right companies and establishing a long-term and successful cooperation.

RFM stands for Recency, Frequency, and Monetary. Recency refers to the date of a customer’s most recent transaction; Frequency captures how often the customer makes

purchases; and Monetary reflects the total value of their spending. The RFM model helps identify which existing customers are most likely to respond positively to a new offer (Sabuncu et al., 2020: 24).

Most of the companies perform the RFM analysis using Microsoft Excel and only few of them are resorting to statistical software such as IBM SPSS Statistics. This is still happening today and is taking considerably more time than using Machine Learning algorithms, not to talk about the possibility of errors or misjudging the potential purchasing power of a customer.

Lately, in different business environments, due to the high increase of customers in companies database and with the introduction of customer Relationship Management (CRM), many businesses decided to flank the RFM analysis with Machine Learning algorithms, such as the K-Means.

Customer segmentation has been performed using recency, frequency, and monetary (RFM) variables, applying K-Means clustering and determining the optimal number of clusters through the elbow method and silhouette score. The analysis identified three cluster groups, classified as best customers, customers at moderate risk of being lost, and average customers (Barus et al., 2023: 298).

The results have clearly demonstrated how the RFM analysis supported by the K-means algorithm, not only significantly shorten the time of processing data, but also, in terms of results, guarantee better clustering with more accurate information which leads to a more focused and better targeted marketing and sales strategy.

One of the main goals of this research is to use the same framework (RFM analysis together with K-means clustering) and applying it to the B2B customer profiling and segmentation in the printing industry.

AI adoption in the printing industry.

The implementation of the AI in the manufacturing is a reality which is rapidly expanding in all the sectors.

Artificial intelligence has emerged as one of the key technological innovations increasingly adopted across the manufacturing sector. The rapid rise of digitalization and the diffusion of

new technological trends have enabled the digital economy to permeate even the largest industrial environments. In the coming years, the role of AI in manufacturing is expected to evolve further, driving improvements in productivity and contributing to higher overall revenue generation (Sukdeo and Mothilall, 2023).

Printing industry worldwide has started considering the usage of AI but only in the manufacturing processes while the implementation of AI on the customer segmentation and profiling is virtually absent and, as a results, also researches about this topic are non-existent. Although the potential of AI within the printing industry appears highly promising, the existing body of research still shows a clear and significant gap concerning its concrete applications and the extent to which these technologies deliver measurable effectiveness (Masod and Zakaria, 2024: 1713).

From a manufacturing perspective, the implementation of AI in the printing manufacturing process is seen as an opportunity to reduce time and costs, which in a declining business as the offset printing, is something paramount.

The integration of Industrial AI into process-optimization efforts has the potential to significantly enhance the profitability of manufacturing operations, while simultaneously supporting greater sustainability and improving overall operational efficiency (Peres et al., 2020).

Is evident that the business owners of printing companies are starting to see and appreciate the implementation of the AI in their operations (specially manufacturing) while are still reluctant to implement AI in their customer segmentation and profiling, considering the traditional methods (mainly the RFM analysis) still effective. This topic offers this research a unique opportunity to demonstrate that the usage of AI and Machine Learning algorithms can significantly enhance the accuracy of the customer segmentation.

As AI-based profiling gains traction in traditional industries, it also raises important ethical and legal considerations, particularly around data privacy and automated decision-making.

Ethical considerations surrounding ML-driven profiling.

Ethical considerations are paramount and represent an important topic when it comes to implementing AI to customer profiling and segmentation.

For that topic is important to introduce the European Union's General Data Protection Regulation (GDPR in this report). This regulation, in force since May 2018, has the main focus on controlling that private individuals and businesses protect the personal information. GDPR marks a major step in the global recognition of personal data's value. While the information economy has existed for years, the true worth of personal information has only recently become clear. Cyber-theft exposes individuals to serious risks, and advanced data-analytics now allow organisations to track and predict individual behaviour, often feeding directly into automated decision-making (IT Governance, 2019).

For this research, focused on the usage of Machine Learning algorithms for customer profiling and segmentation in the B2B printing business, there could be already a problem with the GDPR. The aim of this implementation is to create an automated process to profile and segment the customer base but that could be in conflict with the article 22 of the GDPR, which clearly states that "*the data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her*" (European Parliament and Council of the European Union, 2016).

There are also few other ethical issue to be taken into consideration. In B2B environment, sensible data has to be considered for the segmentation and profiling, such as financial statements, end users pricing structure, volume of sales, margin on sales and pocket margin.

There is also the possibility of biases, for example preferring big organizations rather than smaller or newer ones despite good financial and operational status.

This research directly addresses those gaps by conducting a comparative evaluation of ML-based segmentation against traditional RFM methods within the B2B printing business, with the objective of contributing to both academic knowledge and practical industry advancement.

METHODOLOGY / RESEARCH DESIGN

For this research the philosophy to be used is the positivism and the reason behind this choice is that there will be an objective comparison between the results of the traditional RFM method and the results of the RFM method supported by Machine Learning algorithm. Positivism reflects the philosophical orientation typically associated with natural scientists, emphasising the study of an observable social reality in order to derive law-like generalisations. It is grounded in the belief that such an approach can deliver clear, unambiguous, and reliably accurate knowledge (Saunders et al., 2023: 145).

The approach that will be used is a deductive approach since it will be empirically tested the effectiveness of the new machine learning based approach in profiling and segmenting customers compared with the traditional RFM method.

In fact, if a research starts with theory, developed from a reading of the academic literature, and a research strategy is designed to test the theory, it can be called a deductive approach (Saunders et al., 2023: 154).

This deductive approach will help to compare both the customer profiling and segmenting methods scientifically and, based on the results will determine how effective is the method including the implementation of machine learning algorithms. Therefore the purpose of the research will be based on an evaluative study, in fact evaluative studies within business and management typically focus on determining how effective a particular organisational or business strategy, policy, programme, initiative, or operational process has been in practice (Saunders et al., 2023: 181).

Methodology to be used is the mixed method approach: primarily quantitative to show numerical evidence of the comparison of the two different customer profiling and segmenting techniques and qualitative to support numerical analysis with interpretation to give a logical explanation of the results.

Mixed-methods research design has become one of the fastest-growing approaches in contemporary methodology. Its purpose and advantages are straightforward: draw on the

strengths of both qualitative (QL) and quantitative (QN) methods and integrate them (Bergman, 2008).

The two different customer profiling and segmenting techniques will be tested using secondary data. The choice of secondary data is based on the fact that at the moment there are no research on the B2B offset printing business (mainly on the customer profiling and segmentation) but there are many databases of printing companies working in B2B environment with manufacturing companies. Those databases contain financial information and selling numbers, among other information.

In some cases, using secondary data to address your research questions can save significant time, money, and the effort required for a new primary data collection. Although secondary qualitative and quantitative data were gathered for different purposes, certain elements may still help answer your research questions, including those in mixed-methods studies. (Watkins, 2022: 3).

First step for the practical comparison between the two profiling methods will be use a database containing data of B2B customers and their transactions, including size of customer, type of products purchased, frequency and volume of purchases, volume and value of external sales and payment terms. Initially the idea will be to use an already available database (to be selected in Kaggle).

After performing a thoughtful data cleaning (missing value, duplicated records and wrongly filled fields), the correct data will be standardized to ensure the machine learning algorithm together with the RFM method to optimally perform.

First test will be done on with the traditional RFM method, using a basic Microsoft Excel sheet (as done in reality in many B2B printing businesses) and, assigning a score from one to five (one being the lowest and five being the highest) to the R (Recency), F (Frequency) and M (Monetary). The combination of the score will place each customer in a dedicated segment where a customer scoring a triple five will be considered a “champion customer” and where a customer scoring a triple one will be considered a “lost customer”.

Not only the quality of the results will be recorded but also the time to perform the overall process from start to finish.

Second test will be performed with RFM method supported by K-means machine learning algorithm. The main aim of this algorithm is to partition a dataset into k clusters by reducing the distance among the data points (Solehhuddin et al., 2025: 164).

Elbow method and Silhouette analysis will be then used to determine the perfect cluster.

For this method, the programming language Python will be used.

The usage of Phyton will also allow to include visualisation and graphics to support the qualitative explanation of the results. For this purpose, the library “Matplotlib” will be included in the code.

Matplotlib remains the most widely used general-purpose visualization library, primarily designed for producing static, publication-quality 2D and 3D plots (Johansson, 2019).

At the end there will be a comparison of the results of the segmentation after applying different methods to same dataset, same RFM features and same metrics.

Time efficiency will be also compared as previously stated.

The comparison will finally indicate if the RFM method supported by machine learning algorithm (K-means in this case) is more effective and precise than the traditional RFM method alone.

ETHICAL CONSIDERATIONS AND RISK ASSESMENT

Ethical considerations are paramount within any research involving data processing and, in this case, automated decision-making. Ethical topics have been already discussed in the literature review section, but it is worth to deepen the possible incompatibility between the GDPR regulation and the machine learning customer profiling and segmentation method.

One main point needs to be clarified: GDPR article 22 and Machine Learning automated customer profiling and segmentation tool. As discussed previously, this is one of the sensible parts of the research. Article 22 of the GDPR establishes that individuals have the right not to be subject to decisions based solely on automated processing, including profiling (European Parliament and Council of the European Union, 2016) and the RFM method supported by K-means machine learning algorithm would automatically process the data available and present a customer profiling and segmentation. Nevertheless, the main scope of this experiment is to verify the similarity of results between the traditional RFM method (with excel spreadsheet) and the new RFM method supported by machine learning algorithms (K-means).

When this new method would be introduced in a real work scenario this topic would need to ensure compliance with the article 22 of the GDPR, particularly by maintaining human oversight in the decision-making process.

Risk assessment is another important topic to be considered. Research may involve gathering data in locations that are classified as high-risk or potentially unsafe, which introduces additional ethical and logistical considerations. Many universities now require students to complete a formal risk assessment for such projects, outlining the specific hazards involved and detailing the concrete measures that will be taken to minimise exposure to those risks (Saunders et al., 2023: 214).

In the practical case of this research, the risks are minimal due to the fact that there are no anticipated risks to participants, as the study does not involve any form of human subject engagement. Risks to the researcher remain minimal, given that the project is entirely desktop-based and relies solely on publicly accessible datasets. Risks to the university are

similarly low, provided that the research follows established ethical standards and that all findings are presented with transparency and integrity. All data will be stored securely on password-protected devices, and any methodological constraints or limitations will be openly and accurately acknowledged.

A complete ethical approval application will be submitted together with this research approval in order to be able to start the research having any possible ethical issues and risks assessed and minimized.

DESCRIPTION OF ARTIFACT

There will be only one artifact created for this research and will be a machine learning-based customer segmentation model developed using the Python programming language. The Python code will be written using Anaconda tool for the main reason that Anaconda Code enables local Python/R execution within Excel, with flexible package management and all code and data kept securely in your workbook (Anaconda Inc., 2026).

The model applies K-means clustering to B2B transactional data pre-processed with the RFM (Recency, Frequency, Monetary) framework, enabling direct comparison against traditional RFM-based segmentation.

The model will use three main libraries: *Pandas*, which offers data structures and tools for fast, convenient data cleaning and analysis in Python (McKinney, 2022), *Scikit-learn* which will be used to for the implementation of K-means and the evaluation metrics (Elbow method and Silhouette analysis) and, finally, *Matplotlib*, which has been already discussed in the methodology and research design section of this proposal.

The purpose of this artefact is to serve as the practical foundation for answering the research question. By applying both methods to the same dataset using the same features and metrics, the artefact will generate the empirical evidence needed to evaluate how effectively machine learning techniques classify and segment B2B customers compared with traditional profiling methods. The artefact is also designed to be replicable, allowing printing industry authorised personnels to adapt the framework to their own customer datasets in the future.

For the traditional RFM method, an Excel spreadsheet alone will be sufficient to construct the relevant Pivot Tables and calculate the RFM metrics, offering a practical and accessible baseline against which the K-means clustering results can later be compared, using the same dataset.

TIMELINE FOR PROPOSED ACTIVITIES

Following the approval of the research proposal, the research project is planned over a 22-week period, with an estimated start in May 2026 and conclude by October 2026 (approximately twenty-two weeks after proposal acceptance and approval).

The project will be divided into six separated working packages (WP1 to WP6) and the visualisation of the timeline of the research project can be found in the Gantt Chart available in Appendix A.

Here a brief description of the six working packages:

WP1 = this working package will extensively cover the Literature Review (although a basic Literature Review has been developed also in this research proposal). This extended review will deepen the points highlighted in this proposal and add more literature supporting the subjects already discussed. Duration expected: approximately five weeks.

WP2 = a technical working package which will include the creation of the B2B dataset, cleaning and normalising data and preparation for the RFM traditional model.

Duration expected: approximately three weeks.

WP3 = traditional RFM segmentation applied to the dataset. Analysis and recording of the results and storing of the same. Duration expected: approximately two weeks.

WP4 = the core of this research. Preparation of the machine learning-based customer segmentation model, running the model with the dataset, analysing and storing results and creating visualisations. Duration expected: approximately four weeks.

WP5 = comparative evaluation and analysis of both the results and benchmarks of the same plus qualitative insights and visualisation to be prepared and presented

Duration expected: approximately four weeks.

WP6 = final writing, review, and submission. This includes drafting the dissertation, incorporating feedback, final revisions, and submission. Duration expected: approximately seven weeks. The last two weeks have been allocated in case of rewriting or amendments needed.

A formal ethical approval will be submitted along with this proposal.

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APPENDIX A.

Gantt Chart Research Project. Duration Expected = 22 weeks.

Approximate start: May 2026
Approximate end: October 2026

Research Project Gantt Chart — 22 Weeks

Work Package	May 2026		Jun 2026		Jul 2026		Aug 2026		Sep 2026		Oct 2026
	W1-2	W3-4	W5-6	W7-8	W9-10	W11-12	W13-14	W15-16	W17-18	W19-20	W21-22
WP1: Literature Review & Theoretical Framework											
WP2: Data Collection & Preprocessing											
WP3: Traditional RFM Segmentation											
WP4: ML Model Development & Application											
WP5: Comparative Evaluation & Analysis											
WP6: Writing, Review & Submission											

Legend

	Literature Review
	Data & Preprocessing
	Traditional RFM
	ML Development
	Evaluation
	Writing & Submission